



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



analyze and reverse engineer the sample, enrich with OSIN and build a M...

Research Report

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1. Executive Summary

Lost Boys Cyber assessed the submitted Windows executable as a HelloKitty / Kitty ransomware sample with high confidence. The sample is a 32-bit Windows console PE whose static traits align with public HelloKitty reporting and local detections: ClamAV flags it as `Win.Ransomware.Kitty-9822510-0`, the binary creates or checks the `HelloKittyMutex` mutex, embeds a Tor payment portal, drops `read\_me\_lkdtt.txt`, appends `.crypted`, enumerates local drives and network shares, queries WMI shadow copies, and contains AES/NTRU cryptographic routines.

The sample should be treated as destructive ransomware. It was not detonated on the analyst host because its observable capabilities include recursive file encryption, recycle-bin wiping, shadow-copy handling, network-share enumeration, and self-deletion logic. Dynamic observations in this report are limited to open-source sandbox references and are clearly labeled as OSINT rather than locally reproduced behavior.

Field	Assessment
Malware family	HelloKitty / Kitty ransomware; related reporting also uses DeathRansom lineage terminology
Primary impact	File encryption and extortion via Tor payment portal
Confidence	High for family and capability classification; medium for delivery vector because no phishing/lateral-movement artifact accompanied the sample
Most important defensive action	Isolate any host that executed the sample; collect volatile evidence; hunt for `HelloKittyMutex`, `.crypted`, `read_me_lkdtt.txt`, Tor onion note content, WMI shadow-copy activity, and mass file rewrite patterns
Analyst handling note	Do not execute on analyst workstation or any host with mapped drives; run only in an isolated malware sandbox with disposable storage

2. Intelligence Requirement

This analysis answers the following intelligence requirement: determine what the submitted sample is, what host and enterprise behaviors defenders should expect, what indicators can be used for containment and hunting, and how Lost Boys Cyber should communicate validated findings in a malware-analysis report suitable for portal rendering.

Priority	Requirement	Answer
PIR-1	Is the submitted file malicious?	Yes. Static triage, local AV, imports, strings, and code references indicate destructive ransomware.
PIR-2	What family or cluster is most likely?	HelloKitty / Kitty ransomware, with public reporting tying similar artifacts to DeathRansom lineage discussions.
PIR-3	What does the malware do on a host?	Prevents duplicate execution, prepares a ransom note, encrypts files, appends `.crypted`, searches drives/folders/network shares, handles shadow copies, empties recycle bin, and may self-delete.
PIR-4	What detections and hunts should be deployed?	File, process, WMI, mutex, string/hash, YARA, Sigma, KQL, and SPL coverage

		are provided below.
PIR-5	What evidence is incomplete?	No local detonation telemetry, memory image, victim host triage package, packet capture, or lateral-movement context was provided.

3. Source Review and Confidence

Source	Evidence Used	Confidence	Notes
Submitted evidence archive	Password-protected 7z containing one PE32 executable	High	Archive password `infected`; extracted file SHA-256 matches archive filename.
Local static analysis	`file`, `exiftool`, `rabin2`, `strings`, FLOSS, capa, radare2, ClamAV	High	Multiple independent tools support ransomware classification and HelloKitty family attribution.
OSINT: ANY.RUN search result	Public sandbox result for the same SHA-256 reports process chain involving the sample, `cmd.exe`, `ping.exe`, and `notepad.exe`	Medium	Search-result level evidence only; full sandbox extraction was not available in this environment.
OSINT: Malwarebytes HelloKitty reporting	Describes HelloKitty first seen in November 2020, named for `HelloKittyMutex`, associated ransom note names including `read_me_lkdtt.txt`, `.crypted` extension, AES-128 + NTRU on Windows, and likely delivery through phishing or secondary infection	High	Public reporting matches multiple strings and code artifacts present in this sample.
OSINT: BleepingComputer support-topic search result	Public forum search result references `HelloKittyMutex` and `read_me_lkdtt.txt` in HelloKitty support context	Medium	Page retrieval was blocked by HTTP 403; search result still corroborates public indicator names.

Analytic confidence is high that the sample is HelloKitty ransomware. Confidence is lower for intrusion vector, victimology, and post-compromise staging because the provided evidence contains only the ransomware executable and not the intrusion chain.

Sample and File Metadata

Field	Value
Submitted archive	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e.7z`
Archive SHA-256	`f705d8f19738a4469bf2a014a0150db11353401769829e2543494d3b296cc7f6`
Extracted filename	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`

Extracted MD5	`8e4a887acab5f9475c5fa9a26fb9e720`
Extracted SHA-1	`3294a12a583d2634f6e3d1232052dfe0cd51a44a`
Extracted SHA-256	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`
Size	183,296 bytes
Type	PE32 executable for Microsoft Windows, Intel i386, console subsystem
Linker / compiler indicator	Linker version 14.14; C/C++ runtime artifacts present
PE timestamp	2020-11-12 23:24:34 UTC
Entry point	`0x10783` (image-relative, exiftool)
Sections	`.text`, `.rdata`, `.data`, `.reloc`
Packed?	UPX test reported not packed by UPX; no high-confidence generic packer label observed
Local AV	ClamAV: `Win.Ransomware.Kitty-9822510-0 FOUND`

4. Key Findings

Finding	Evidence	Defensive Implication	Confidence
The sample is HelloKitty ransomware.	`HelloKittyMutex`, `read_me_lkdtt.txt`, `.crypted`, Tor onion ransom portal, ClamAV Kitty detection, and OSINT family overlap.	Treat host execution as an active ransomware incident requiring isolation and enterprise-wide hunt.	High
The sample prevents duplicate execution through a named mutex.	radare2 shows `OpenMutexW` and `CreateMutexW` references to `HelloKittyMutex` in `main` around `0x409650` and `0x409674`.	Mutex telemetry can support EDR hunting and sandbox family confirmation.	High
The sample embeds a full ransom note and Tor payment portal.	FLOSS/rabin2 recovered note text beginning `Hello dear user. Your files have been encrypted` and onion URL `http://6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion`.	Ransom-note creation is a late-stage impact signal; filesystem detections should trigger incident response immediately.	High
The sample uses strong cryptographic components consistent with HelloKitty reporting.	capa identified manually built AES constants; strings include NTRU functions such as `ntru_crypto_ntru_encrypt`, `ntru_crypto_ntru_encrypt_key_gen`, and DRBG routines. Malwarebytes reporting states Windows HelloKitty uses	Recovery should not rely on cryptographic weakness; prioritize backups, containment, and legal/IR process.	High

	AES-128 + NTRU.		
The sample searches local drives, command-line supplied paths, and network resources.	Imports and capa show `GetLogicalDrives`, recursive file enumeration, `WNetOpenEnumW`, `WNetEnumResourceW`, and command-line parsing; strings include `-path`, `SearchFolder`, and `DriveSearchThread`.	Mapped drives and reachable shares are at risk; isolate endpoints and disable compromised credentials.	High
The sample attempts destructive host preparation.	Strings/imports include `SHEmptyRecycleBinA`, WMI query `select * from Win32_ShadowCopy`, `Win32_ShadowCopy.ID=%s`, Restart Manager functions, file attribute manipulation, and self-deletion command `C ping 127.0.0.1 & del %s`.	Monitor for backup tampering and recovery degradation; collect shadow-copy state and event logs early.	High
The sample is multi-threaded and optimized for mass file processing.	capa identified `CreateIoCompletionPort`, thread creation, and file read/write loops; strings include `Number of threads %u`, `read=%u kbytes, write=%u kbytes`, `TotalEncrypted=%u`, and `Total time of encryption`.	Endpoint file I/O telemetry should show rapid multi-file read/write/rename activity.	High

Static Analysis Highlights

The binary imports or dynamically references APIs associated with encryption, file traversal, system modification, and victim interaction:

Capability	APIs / Strings	Interpretation
Cryptographic setup	`CryptAcquireContextW`, `CryptGenRandom`, `ntnu_crypto_*`, AES constants	Random key generation and asymmetric wrapping consistent with ransomware encryption design.
File encryption loop	`CreateFileW`, `ReadFile`, `WriteFile`, `MoveFileW`, `SetFileAttributesW`, `.crypted`	Reads original content, writes encrypted content, and renames or marks files.
Parallel processing	`CreateThread`, `CreateIoCompletionPort`, `GetQueuedCompletionStatus`, `PostQueuedCompletionStatus`	Uses worker threads and IOCP-like design to process files at scale.
Drive/share discovery	`GetLogicalDrives`, `GetDriveTypeW`, `WNetOpenEnumW`, `WNetEnumResourceW`	Targets local volumes and network resources.
Recovery interference	`select * from Win32_ShadowCopy`, `SHEmptyRecycleBinA`, Restart Manager strings	Attempts to reduce easy restore paths and unlock files.
User notification	`read_me_lkdtt.txt`, Tor onion URL,	Drops and likely opens a ransom note.

	`ShellExecuteW`, `notepad.exe` observed in OSINT sandbox search result	
Anti-forensics	`/C ping 127.0.0.1 & del %s`, capa self-delete	Delayed self-deletion after execution.

Indicators and Observables

Type	Indicator
SHA-256	`c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`
SHA-1	`3294a12a583d2634f6e3d1232052dfe0cd51a44a`
MD5	`8e4a887acab5f9475c5fa9a26fb9e720`
Archive SHA-256	`f705d8f19738a4469bf2a014a0150db11353401769829e2543494d3b296cc7f6`
Mutex	`HelloKittyMutex`
Ransom note	`read_me_lkdtt.txt`
Encrypted file extension	`.crypted`
Tor/onion portal	`http://6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrda.d.onion`
WMI query	`select * from Win32_ShadowCopy`
WMI object format	`Win32_ShadowCopy.ID=%s`
Self-delete command pattern	`/C ping 127.0.0.1 & del %s`
AV family label	`Win.Ransomware.Kitty-9822510-0`

5. Threat Context

HelloKitty, also referred to as Kitty ransomware in public reporting, emerged around November 2020 and became widely discussed after reporting connected it to the CD Projekt Red ransomware incident. Public reporting states that the family was named after its `HelloKittyMutex` mutex and that some researchers discuss it in relation to DeathRansom lineage. Malwarebytes reporting describes Windows HelloKitty as using AES-128 with NTRU, appending `.kitty` or `.crypted`, and dropping ransom notes named `read\_me\_lkdtt.txt` or `read\_me\_unlock.txt`.

This submitted sample aligns tightly with those public traits. The binary's embedded `HelloKittyMutex`, `.crypted` extension, `read\_me\_lkdtt.txt` note name, NTRU symbols, AES capability, and Tor note template make the attribution stronger than a generic ransomware classification. The sample also includes operational behaviors common to human-operated ransomware payloads: recursive enumeration, network-share traversal, process/file unlocking support through Restart Manager, and recovery interference through shadow-copy and recycle-bin handling.

Likely execution scenario based on the sample and OSINT:

Phase	Assessment
Initial access	Not established from the sample. Public HelloKitty reporting cites

	phishing or secondary deployment after an initial compromise.
Payload deployment	Operator or prior malware deploys the ransomware on a Windows host, potentially with a command-line <code>`-path`</code> argument to scope encryption.
Execution control	Sample checks <code>`HelloKittyMutex`</code> to avoid duplicate runs, sets process error mode, and initializes COM/WMI.
Discovery and targeting	Enumerates drives, folders, and network shares; skips common system paths and files such as <code>`windows`</code> , <code>`program files`</code> , <code>`programdata`</code> , <code>`\$recycle.bin`</code> , <code>`ntuser.dat`</code> , <code>`pagefile.sys`</code> , and <code>`desktop.ini`</code> .
Impact	Encrypts eligible files, appends <code>`.crypted`</code> , writes <code>`read_me_lkdt.txt`</code> , directs victims to a Tor portal, and may open the note.
Anti-recovery / cleanup	Uses WMI shadow-copy artifacts, empties recycle bin, and contains self-delete logic.

MITRE ATT&CK Mapping

Tactic	Technique	Evidence
Execution	T1059.003 - Windows Command Shell	Self-delete command uses <code>`cmd.exe /C ping 127.0.0.1 & del %s`</code> ; OSINT sandbox result references <code>`cmd.exe`</code> and <code>`ping.exe`</code> .
Defense Evasion	T1070.004 - File Deletion	capa self-delete finding and delayed deletion command.
Defense Evasion	T1070 - Indicator Removal	Recycle-bin emptying and self-deletion reduce local recovery/evidence.
Discovery	T1083 - File and Directory Discovery	Recursive file enumeration and <code>`SearchFolder`/`DriveSearchThread`</code> strings.
Discovery	T1135 - Network Share Discovery	<code>`WNetOpenEnumW`</code> , <code>`WNetEnumResourceW`</code> , and <code>`WNetCloseEnum`</code> imports.
Discovery	T1120 - Peripheral Device Discovery	Drive enumeration through <code>`GetLogicalDrives`</code> and <code>`GetDriveTypeW`</code> .
Impact	T1486 - Data Encrypted for Impact	Ransom note, <code>`.crypted`</code> , AES/NTRU functions, and file read/write/rename behavior.
Impact	T1490 - Inhibit System Recovery	WMI <code>`Win32_ShadowCopy`</code> query and recycle-bin clearing.
Impact	T1489 - Service Stop	Public HelloKitty reporting notes service/process termination; this sample imports Restart Manager functions, but a local termination list was not recovered.
Command and Control / Exfiltration	T1090.003 - Multi-hop Proxy / Tor	Ransom note instructs Tor browser use and

Adjacent		embeds an `.onion` URL.
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Detection Coverage Matrix

Detection Surface	Recommended Logic	Expected Signal	Priority
File hash	Block and alert on SHA-256/MD5/SHA-1 listed above	Known sample execution or staging	Critical
Mutex	Alert on process creation/use of `HelloKittyMutex` where telemetry exposes mutex objects	Family-specific execution marker	High
Ransom note	Creation of `read_me_lkdtt.txt`	Post-encryption or active encryption signal	Critical
File extension	Rapid creation/rename to `*.crypted`	Encryption impact	Critical
WMI	`Win32_ShadowCopy` query or deletion by unusual process	Recovery inhibition	High
Network/share enumeration	Non-admin process enumerating shares shortly before mass writes	Ransomware staging	High
Self-delete	`cmd.exe /c ping 127.0.0.1 & del <path>`	Cleanup after payload run	Medium
Tor note content	`.onion` string in new text files or process command context	Victim instruction artifact	Medium

YARA Rule

```
rule LBC_HelloKitty_Ransomware_c7d6719b
{
  meta:
    description = "Detects submitted HelloKitty ransomware sample and close variants with shared note/mutex artifacts"
    author = "Lost Boys Cyber"
    date = "2026-06-17"
    sha256 = "c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e"
    confidence = "high"
  strings:
    $mz = { 4D 5A }
    $mutex = "HelloKittyMutex" wide ascii
    $note_name = "read_me_lkdtt.txt" wide ascii
    $ext1 = ".crypted" wide ascii
    $onion = "6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion" wide ascii
    $shadow = "select * from Win32_ShadowCopy" wide ascii
    $ntru1 = "ntru_crypto_ntru_encrypt" ascii
    $ntru2 = "ntru_crypto_ntru_encrypt_keygen" ascii
    $ransom = "Your files have been encrypted" wide ascii
  condition:
    $mz at 0 and uint16(0) == 0x5A4D and filesize < 1MB and
    (
      all of ($mutex, $note_name, $ext1, $onion) or
      5 of ($mutex, $note_name, $ext1, $onion, $shadow, $ntru1, $ntru2, $ransom)
    )
}
```

}

Sigma Rule

```

title: HelloKitty Ransomware Note or Encrypted Extension Artifact
id: 817cf7df-6f30-46f5-9c8f-fc57f4b0c7d6
status: experimental
description: Detects file artifacts associated with the analyzed HelloKitty ransomware sample.
author: Lost Boys Cyber
date: 2026-06-17
references:
  - https://www.malwarebytes.com/blog/news/2021/03/hellokitty-when-cyberpunk-met-cy-purr-crime
logsource:
  product: windows
  category: file_event
detection:
  selection_note:
    TargetFilename|endswith: '\read_me_lkdtt.txt'
  selection_ext:
    TargetFilename|endswith: '.crypted'
  condition: selection_note or selection_ext
fields:
  - Image
  - TargetFilename
  - User
  - Computer
falsepositives:
  - Malware-analysis lab activity
  - Test files intentionally named after HelloKitty artifacts
level: critical
tags:
  - attack.impact
  - attack.t1486

```

Microsoft Defender KQL Hunts

```

// Known sample staging or execution by hash
DeviceFileEvents
| where SHA256 =~ "c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e"
  or MD5 =~ "8e4a887acab5f9475c5fa9a26fb9e720"
  or SHA1 =~ "3294a12a583d2634f6e3d1232052dfe0cd51a44a"
| project Timestamp, DeviceName, FolderPath, FileName, SHA256, InitiatingProcessFileName,
InitiatingProcessCommandLine

// Ransom note and encrypted-extension burst behavior
let window = 10m;
DeviceFileEvents
| where ActionType in ("FileCreated", "FileRenamed", "FileModified")
| where FileName =~ "read_me_lkdtt.txt" or FolderPath endswith ".crypted" or FileName endswith
".crypted"
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), FileCount=count(),
Paths=make_set(FolderPath, 20)
  by DeviceName, InitiatingProcessFileName, InitiatingProcessSHA256, bin(Timestamp, window)
| where FileCount >= 3 or Paths has_any ("read_me_lkdtt.txt", ".crypted")
| order by LastSeen desc

// Shadow copy WMI activity by unusual process
DeviceProcessEvents
| where ProcessCommandLine has_any ("Win32_ShadowCopy", "vssadmin", "wmic shadowcopy", "Delete
Shadows")
  or InitiatingProcessCommandLine has "Win32_ShadowCopy"
| project Timestamp, DeviceName, FileName, ProcessCommandLine, InitiatingProcessFileName,
InitiatingProcessCommandLine, AccountName

// Self-delete pattern used by the sample
DeviceProcessEvents
| where FileName =~ "cmd.exe"
| where ProcessCommandLine has_all ("ping", "127.0.0.1", "del")
| project Timestamp, DeviceName, ProcessCommandLine, InitiatingProcessFileName,

```

InitiatingProcessCommandLine, AccountName

Splunk Hunts

```

index=windows (sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" OR
sourcetype="WinEventLog:Security")
("read_me_lkdtt.txt" OR ".crypted" OR "HelloKittyMutex" OR "Win32_ShadowCopy")
| table _time host user Image CommandLine TargetFilename EventCode
| sort - _time

index=windows EventCode=1 Image="*\\cmd.exe" CommandLine="*ping*127.0.0.1*del*"
| table _time host user ParentImage Image CommandLine
| sort - _time

```

Triage Checklist for Suspected Hosts

Step	Action	Rationale
1	Isolate host from network, including VPN and mapped shares.	Prevent encryption from spreading to reachable shares.
2	Preserve memory and running process list if encryption is still active.	May capture keys, active handles, or process lineage.
3	Collect `\$MFT`, USN Journal, Sysmon/EDR telemetry, PowerShell/WMI logs, and Windows Event Logs.	Reconstruct file rename/write timeline and recovery-inhibition actions.
4	Search for `read_me_lkdtt.txt`, `.crypted`, and onion note content.	Scope impacted directories.
5	Check VSS/shadow-copy state and backup-console logs.	Determine restore viability.
6	Hunt for credential misuse and prior-stage malware.	HelloKitty is often reported as deployed after phishing or a secondary infection, not necessarily as the first-stage payload.

7. Recommendations

Priority	Recommendation	Owner	Timeline
Critical	Block the submitted hashes across EDR, email gateway, web proxy, and malware repository controls.	Security Engineering	Immediate
Critical	Add detection for `read_me_lkdtt.txt`, `.crypted`, `HelloKittyMutex`, `Win32_ShadowCopy`, and the Tor onion indicator.	Detection Engineering	Immediate
Critical	If this sample executed in production, isolate affected hosts and disable potentially compromised domain, local admin, and service accounts.	Incident Response / IAM	Immediate
High	Review backup immutability, offline restore points, and shadow-copy telemetry for	Infrastructure / Backup Operations	24 hours

	tampering.		
High	Hunt across file servers and endpoints for rapid file renames/writes and network-share enumeration from unusual processes.	SOC / Threat Hunting	24 hours
High	Confirm EDR visibility for file creation, process command line, WMI, module loads, and network share enumeration.	Security Engineering	72 hours
Medium	Add YARA scanning for malware repositories, file shares, and suspicious executable staging directories.	Malware Defense	72 hours
Medium	Exercise ransomware containment runbooks with a scenario involving mapped drives and shadow-copy tampering.	IR Lead	30 days

Operational notes:

- Do not pay ransom or contact the onion portal without executive, legal, insurer, and law-enforcement coordination.
- Do not delete ransom notes or encrypted files before forensic collection; they may be needed for scoping and recovery decisions.
- Prioritize identity containment. Ransomware payload execution often follows credential theft or interactive intrusion activity.
- Verify backup integrity from a clean restore environment before reconnecting recovered hosts to production.

8. References

- Submitted sample archive: `\home\lost0x01\internal-secops-portal\data/uploads\20260617_183629_893432\c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e.7z``.
- Local analysis artifacts generated in workspace: `\home\lost0x01\hermes\kanban\workspaces\t_36cfe657\static\``.
- Malwarebytes, ``HelloKitty: When Cyberpunk met cy-purr-crime``, 2021-03-18. Retrieved evidence includes family naming from ``HelloKittyMutex``, first-seen timing, likely infection vectors, ``crypted\`.kitty`` extension behavior, ``read_me_lkdtt.txt`` ransom-note naming, and AES/NTRU description.
- ANY.RUN public search result for SHA-256 ``c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e``, indicating sandboxed process chain references to the sample, ``cmd.exe``, ``ping.exe``, and ``notepad.exe``.
- BleepingComputer public search result for HelloKitty support topic referencing ``HelloKittyMutex`` and ``read_me_lkdtt.txt``; direct page retrieval was blocked by HTTP 403 during this analysis.
- Local tool outputs: ClamAV ``Win.Ransomware.Kitty-9822510-0``, capa 9.4.0 capability report, FLOSS 3.1.1 string extraction, radare2 function/string cross references, exiftool PE metadata, and rabin2 imports/sections/strings.